

An Internship Report

Of

Python Programming Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (CSE)



By

Enrollment / Roll Number: 25SCS1003004407

Name of Student: ANANT CHAUDHARY

Semester: 3rd Semester | Section: 2CSE36

Session: 2025–29

1. Candidate's Declaration

I, Anant Chaudhary, hereby declare that this internship report titled “CodeAlpha Python Programming Internship” has been prepared by me to fulfil the academic requirements of the Bachelor of Technology in Computer Science and Engineering. The report documents the work completed during my one-month virtual internship at CodeAlpha from 1st July 2026 to 30th July 2026. The project descriptions are based on my submitted source files and documentation. I take responsibility for the originality and accuracy of the work presented.

Student Name: Anant Chaudhary

Roll No.: 25SCS1003004407

2. Acknowledgement

I express my sincere gratitude to CodeAlpha for providing me the opportunity to participate in its Virtual Internship Program in Python Programming. The internship gave me practical exposure to writing Python programs, applying programming fundamentals, handling user input, validating data, organizing program logic, and producing working console applications.

I am thankful to the CodeAlpha team for the learning environment and guidance provided during the internship. I also thank the faculty and the School of Computer Science and Engineering, IILM University, for encouraging practical learning. Finally, I am grateful to my parents and everyone who supported me throughout the internship.

Student Name: Anant Chaudhary

3. Internship Completion Certificate

The CodeAlpha Certificate of Completion identifies Anant Chaudhary (Student ID CA/DF1/182227) as an active participant in the CodeAlpha Virtual Internship Program in Python Programming. It records the internship period as 1st July 2026 to 30th July 2026 and the date of issue as 31st July 2026.

The original certificate is supplied separately as “Internship Certificate.pdf”. It is kept unchanged for submission authenticity.

Certificate Detail	Recorded Information
Candidate	Anant Chaudhary
Student ID	CA/DF1/182227
Program	CodeAlpha Virtual Internship Program
Domain	Python Programming
Duration	1st July 2026 to 30th July 2026
Date of Issue	31st July 2026

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5. Project Description

The internship work consisted of three Python programming projects designed to strengthen programming fundamentals through practical implementation. The projects cover data processing and file handling, rule-based interaction, and game logic. All three applications are console-oriented and use Python 3.

4.1 Introduction

The CodeAlpha Python Programming Internship provided an opportunity to apply core Python concepts to working programs. The tasks focused on building simple applications from defined requirements and validating their behaviour through interactive input. The completed projects demonstrate practice with dictionaries and file handling, conditional logic and loops, and state management through lists and strings.

4.2 Organization Profile

The supplied CodeAlpha completion documents identify the program as a Virtual Internship Program in Python Programming. The certificate confirms participation from 1st July 2026 to 30th July 2026, and the Letter of Recommendation confirms successful completion of the one-month internship and describes analytical skills, adaptability to emerging technologies, productivity, assistance and collaboration.

4.3 Problem Statement

The internship tasks required small, functional Python applications that solve clearly defined interactive problems. The main challenge was to translate requirements into reliable program flow while validating user input and producing understandable output. The portfolio task calculates and summarizes investment value; the chatbot maps user messages to predefined responses; and Hangman maintains a hidden word, processes guesses and determines win or loss.

4.4 Project Objectives

- Strengthen Python programming fundamentals through hands-on implementation.
- Practice functions, loops, conditionals, collections, string handling and user input.
- Implement input validation and simple error handling.
- Work with file handling by saving a portfolio summary to a text file.
- Develop simple interactive applications with clear terminal-based feedback.
- Improve problem decomposition and documentation skills.

4.5 Scope of the Project

The scope covers three standalone console applications written in Python 3. The Stock Portfolio Tracker uses a fixed set of predefined stock prices and saves a summary to a local text file. The Basic Chatbot uses predefined conversational rules. The Hangman Game uses a small predefined word list and a six-error limit. These are educational task projects rather than production financial, AI or gaming systems.

4.6 Technologies and Tools Used

Project	Technology	Key Concepts
Stock Portfolio Tracker	Python 3	Dictionary, Loops, Conditional statements, User input, File handling
Basic Chatbot	Python 3	Functions, if-elif-else, while loop, User input, String handling
Hangman Game	Python 3	random module, Lists, Strings, while loop, if-else, User input

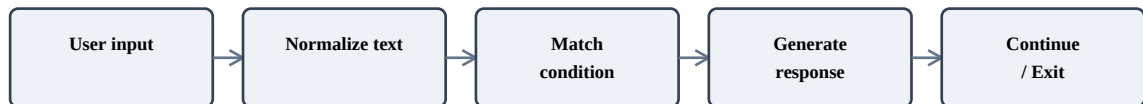
4.7 System Architecture

Each supplied application follows a lightweight input–processing–output architecture. Program state is held in Python variables and collections. The portfolio application additionally writes a local text summary.

Stock Portfolio Tracker



Basic Chatbot



Hangman Game



4.8 Methodology

- Requirement understanding: identify inputs, outputs and rules.
- Design: select suitable Python data structures and control flow.
- Implementation: develop the terminal application using Python 3.
- Validation: check invalid symbols, quantities, commands and guesses.
- Output generation: display readable results and save portfolio data where required.
- Review: compare program behaviour with the task requirements.

4.9 Expected Outcomes

- Functional Stock Portfolio Tracker with calculated value and text-file output.
- Functional rule-based chatbot for its predefined conversational inputs.

- Functional Hangman game with random selection, validation and win/loss handling.
- Improved understanding of Python fundamentals and console application development.
- Practical experience translating task statements into executable logic.

4.10 Certificates of Completion and Communication Proof

The following screenshots are included as communication proof of internship completion. They have been cropped to retain the relevant CodeAlpha/Gmail content while removing browser controls and unrelated side panels.

4.10.1 Completion Email Screenshot

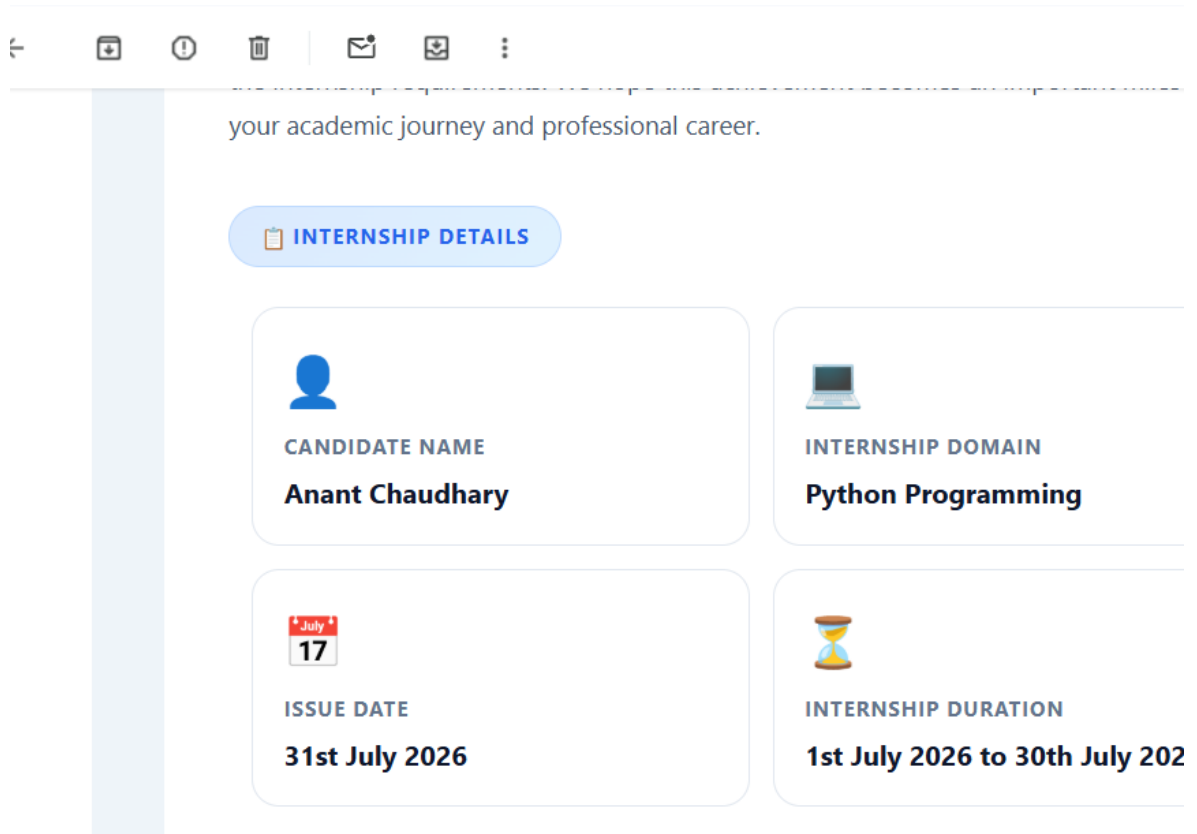


Figure 4.10.1: CodeAlpha internship completion email showing the candidate, Python Programming domain, internship duration and issue date.

4.10.2 Completion Package / Documents Included Screenshot

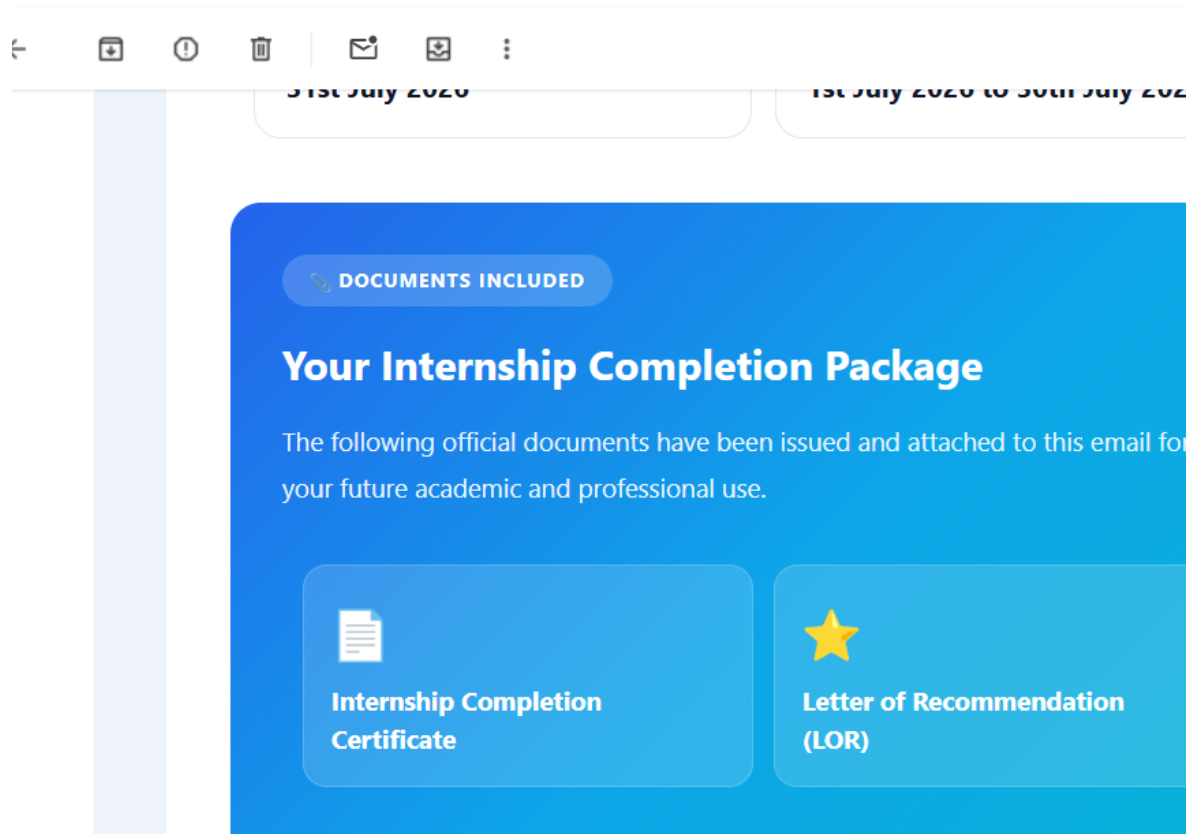


Figure 4.10.2: CodeAlpha completion package email showing the Internship Completion Certificate and Letter of Recommendation documents.

6. Project-wise Implementation

6.1 Stock Portfolio Tracker

Project Overview

A console-based Python application that accepts stock symbols and quantities, calculates portfolio value using predefined stock prices, displays a portfolio summary, and saves the result to a text file.

Technology

Python 3

Key Features

- Predefined prices for AAPL, TSLA, GOOGL, MSFT and AMZN.
- Validates stock symbols.
- Accepts quantities and handles invalid numeric input.
- Calculates per-stock and total investment value.
- Saves the portfolio summary to portfolio.txt.

Implementation and Logic

The program uses a dictionary of predefined stock prices and a portfolio dictionary for quantities. It repeatedly accepts a symbol, validates it, reads quantity inside a try/except block, and accumulates holdings. At the end it calculates values, prints a summary and writes the result to portfolio.txt.

Process Flow



6.2 Basic Chatbot

Project Overview

A simple rule-based Python chatbot that responds to predefined user inputs and supports an interactive terminal conversation.

Technology

Python 3

Key Features

- Interactive command-line conversation.
- Recognizes predefined greetings.
- Answers predefined questions.
- Provides a help command.
- Terminates on bye or exit; unknown inputs receive a fallback response.

Implementation and Logic

The chatbot runs inside a function and uses a while loop. Input is lowercased and stripped before comparison with predefined phrases. Matching branches return fixed responses, help lists supported commands, and bye/exit ends the conversation.

Process Flow



6.3 Hangman Game

Project Overview

A console-based Hangman game in which the program randomly selects a word from a predefined list and the player guesses one letter at a time.

Technology

Python 3

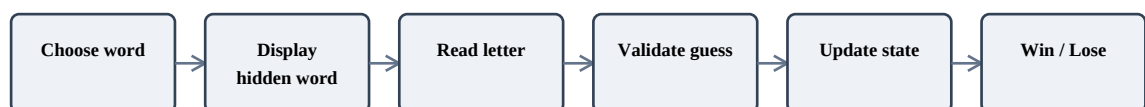
Key Features

- Randomly selects from five predefined words.
- Displays hidden letters using underscores.
- Accepts one alphabetic character per turn.
- Rejects invalid and repeated guesses.
- Allows a maximum of six incorrect guesses.

Implementation and Logic

The program uses `random.choice` to select one word from a predefined list. It stores guessed letters, counts wrong guesses, validates one alphabetic character, rejects duplicate guesses, updates the masked word and ends when the word is revealed or six wrong guesses are reached.

Process Flow



7. Learning Outcomes

- Applied Python syntax and programming fundamentals in multiple practical tasks.
- Used dictionaries and lists to represent application state.
- Used loops and conditional branches to control interactive program flow.
- Implemented input validation and exception handling for selected user-input cases.
- Practiced string normalization and comparison for rule-based interaction.
- Used the random module to introduce word selection in a game.
- Learned basic file handling by creating a persistent text summary.
- Improved problem decomposition, debugging awareness and documentation.

8. Conclusion

The CodeAlpha Python Programming Internship provided practical experience through three focused programming tasks: Stock Portfolio Tracker, Basic Chatbot and Hangman Game. Although the applications are compact, each reinforces an important area of Python programming. Together they demonstrate input handling, control flow, data management, validation, file handling and interactive application design. The internship therefore provided a useful practical foundation for further development in Python and software programming.

The supplied CodeAlpha certificate confirms completion from 1st July 2026 to 30th July 2026, and the communication proof screenshots document the completion package containing the certificate and Letter of Recommendation.

9. Bibliography / References

- CodeAlpha, Certificate of Completion issued to Anant Chaudhary, Student ID CA/DF1/182227, 31st July 2026.
- CodeAlpha, Letter of Recommendation issued to Anant Chaudhary, Student ID CA/DF1/182227, 31st July 2026.
- CodeAlpha internship project source files supplied by the student: Stock Portfolio Tracker, Basic Chatbot and Hangman Game.
- CodeAlpha project README documentation supplied by the student.
- IILM University, School of Computer Science and Engineering, Internship Report Format (updated 2026–27).